

By GROUP 1

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Module 05 Lab - Data Preparation

Objective:

To learn and apply the most common data preparation techniques. Raw data is rarely ready for a machine learning model. This process, also called preprocessing, is one of the most critical steps in the entire machine learning workflow.

In this lab, you will write more of the code.

Read the explanations and then complete the tasks in the code cells.

Part 1: Setup and Initial Look

We will continue using the Titanic dataset because it has the exact problems we need to solve: missing values and non-numeric data.

```
import pandas as pd
import numpy as np

# Load the dataset
df = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv')

# Let's look at the missing values
print("--- Missing Values Before Cleaning ---")
print(df.isnull().sum())

--- Missing Values Before Cleaning ---
PassengerId      0
Survived          0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

Part 2: Handling Missing Values (Imputation)

Concept: Most machine learning models cannot handle missing values (NaN). We must deal with them. Dropping the rows is an option, but you lose data. A better way is **imputation**, which means filling in the missing values with a calculated guess.

Common imputation strategies:

- Mean:** Fill with the average value. Good for normally distributed data.
- Median:** Fill with the middle value. Better for skewed data or data with outliers (like Fare).
- Mode:** Fill with the most frequent value. Used for categorical data.

Task 1: Impute the 'Age' ColumnThe 'Age' column is missing many values. Since age can be skewed (e.g., by a few very old passengers), using the **median** is a robust choice.**Your Task:** Calculate the median of the 'Age' column and use the .fillna() method to replace the missing values.

```
# 1. Calculate the median of the 'Age' column
median_age = df['Age'].median()

# 2. Fill the missing values in 'Age' with the median
df['Age'].fillna(median_age, inplace=True)

# 3. Verify that there are no more missing values in 'Age'
print("Missing values in 'Age' after imputation:")
print(df['Age'].isnull().sum())

Missing values in 'Age' after imputation:
0
/tmp/ipykernel_1849/2401602677.py:5: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on th

df['Age'].fillna(median_age, inplace=True)
```

Part 3: Encoding Categorical Features

Concept: Machine learning models are mathematical, so they need numbers, not text. We need to convert categorical columns (like Sex and Embarked) into a numerical format. The most common method is **One-Hot Encoding**.

One-Hot Encoding takes a column with N categories and turns it into N new columns, each with a 1 or 0.

For example, the Sex column (male, female) becomes two new columns: Sex_male and Sex_female.

Pandas has a convenient function called pd.get_dummies() that does this for us.

Task 2: One-Hot Encode Categorical Columns

Your Task: Use pd.get_dummies() to encode the Sex and Embarked columns. Make sure to drop the original columns after encoding.

```
# --- Task 2: One-Hot Encoding ---

# 1. Use get_dummies to create new columns for 'Sex' and 'Embarked'
encoded_df = pd.get_dummies(
    df,
    columns=['Sex', 'Embarked'],
    drop_first=True
```

```
)  
  
# 2. Display the first few rows of the new DataFrame  
print(encoded_df.head())
```

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Age	SibSp	Parch	\
0	Braund, Mr. Owen Harris	22.0	1	0	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	38.0	1	0	
2	Heikkinen, Miss. Laina	26.0	0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	35.0	1	0	
4	Allen, Mr. William Henry	35.0	0	0	

	Ticket	Fare	Cabin	Sex_male	Embarked_Q	Embarked_S
0	A/5 21171	7.2500	NaN	True	False	True
1	PC 17599	71.2833	C85	False	False	False
2	STON/O2. 3101282	7.9250	NaN	False	False	True
3	113803	53.1000	C123	False	False	True
4	373450	8.0500	NaN	True	False	True

▼ Part 4: Feature Scaling

Concept: Many models are sensitive to the scale of the features. For example, `Age` (from 0-80) and `Fare` (from 0-512) are on very different scales. This can cause the model to incorrectly believe that `Fare` is a more important feature simply because its values are larger.

Feature Scaling solves this by putting all features on a similar scale.

A common method is **Standardization** (`StandardScaler` in scikit-learn), which rescales the data to have a mean of 0 and a standard deviation of 1.

Important: You only scale your numerical features, not your target variable or your newly encoded categorical columns.

▼ Task 3: Scale the 'Age' and 'Fare' Columns
Your Task: Use `StandardScaler` from `sklearn.preprocessing` to scale the 'Age' and 'Fare' columns.

```
from sklearn.preprocessing import StandardScaler  
  
# Create the scaler  
scaler = StandardScaler()  
  
# Scale Age and Fare  
encoded_df[['Age', 'Fare']] = scaler.fit_transform(  
    encoded_df[['Age', 'Fare']]  
)  
  
# Display the results  
print(encoded_df[['Age', 'Fare']].head())
```

	Age	Fare
0	-0.565736	-0.502445
1	0.663861	0.786845
2	-0.258337	-0.488854
3	0.433312	0.420730
4	0.433312	-0.486337

 Knowledge Check

Instructions: Answer the following questions in this markdown cell.

1. **Why is it often better to impute missing values with the median instead of the mean?**

It is often better to use the median because it is less affected by extreme values or outliers. The mean can be pulled higher or lower by unusually large or small values, while the median represents the middle value and is more robust. In this lab, using the median for the Age column helps provide a reasonable replacement for missing ages without being strongly affected by unusually young or old passengers.

2. **Explain in your own words what One-Hot Encoding does and why it is necessary.**

One-Hot Encoding converts categorical values into separate numerical columns that use 0s and 1s. This step allows a machine learning model to work with categories such as male, female, or different embarkation ports without treating the category names as numerical values. It also helps prevent the model from assuming that one category has a higher or lower value than another. For this reason, One-Hot Encoding gives the model a clearer and more appropriate way to use categorical information.

3. **Would you need to apply Feature Scaling to a Decision Tree model?** Why or why not? (Hint: Think about how a Decision Tree makes its splits).

Feature scaling is usually not necessary for a Decision Tree because the model makes splits based on feature values rather than the overall scale of the features. Scaling changes the size of the values, but it does not change the order of the observations or the decisions the tree can make.